

Objective

The objective of this document is to define the end-to-end approach for integrating the Lepton GIS hybrid application—built using React, Vite, deck.gl, Tailwind, and Capacitor—into the HSC native mobile app. This integration should enable HSC users to access Lepton's GIS capabilities within the HSC app.

IT & Build Environment Prerequisites

- **Network Whitelisting (to be handled by IT team)**

The following URLs must be allowlisted to enable dependency installation and native builds:

- <https://registry.npmjs.org/> — npm packages
- <https://maven.google.com/> — Android libraries
- <https://repo1.maven.org/maven2/> — Maven Central
- <https://services.gradle.org/> — Gradle services

- **Versions List:**

Node Version: 22.12.0

Yarn Version: 1.22.22

Rest of the versions in the package.json.

- **TLS / Secure Socket Configuration**

- Ensure **TLS sockets / HTTPS traffic** are enabled and not blocked by corporate network policies, as secure connections are required for dependency resolution, build services, and runtime communications.

- **Build Tooling Configuration**

- **Yarn** must be available in the build environment
- Required **Yarn environment variables** should be added to the **global environment variables** by the IT team

Prerequisites

Before proceeding, ensure that the following prerequisites are met:

- The **Lepton Android codebase**, named **mcsa-gis-android**, exists in the root of the **deal_wf_mcsa-develop**.
- The **mcsa-gis-android** project can be successfully built by running the following commands in sequence:

```
• Cd mcsa-gis-android
  npm install -g yarn (if not installed)
  yarn install
  yarn build
```

```
• npx cap sync
  yarn sync:mcsa-gis-android-sync (this will skip -> Step 9: Copy Web Assets)
```

Exact Steps for integration:

Step 1: Update Root **build.gradle.kts** and **settings.gradle.kts** and **.gitignore**

File:

deal_wf_mcsa-develop/.gitignore

Changes to Apply

```
• # mcsa-gis-android dependencies and build outputs (if mcsa-gis-android is inside integration-main)
• mcsa-gis-android/node_modules/
• mcsa-gis-android/dist/
• mcsa-gis-android/.yarn/
• mcsa-gis-android/.pnp.*
```

File:

deal_wf_mcsa-develop/build.gradle.kts

Changes to Apply

1. **Add the following import at the top of the file:**
 - **import com.android.build.gradle.LibraryExtension**
2. **Apply Capacitor variables file:**
 - Add the following line **after the **plugins {}** block:**
 - **apply(from = "variables.gradle")**
3. **Add a namespace override for the Capacitor file picker plugin:**
 - Append the following **at the bottom of the file:**

- A `subprojects {}` block that sets the `namespace` for the `capacitor-file-picker` module to comply with AGP 8+ requirements.

Reason

- `variables.gradle` provides required SDK versions to all Capacitor modules.
- During the **initial attempt to integrate the `capacitor-file-picker` plugin**, a **“Namespace not specified”** build error was encountered when using **Android Gradle Plugin (AGP) 8+**.
The namespace override was added to resolve this issue.

File:

`deal_wf_mcsa-develop/settings.gradle.kts`

Update:

```
• pluginManagement {
•
•     repositories {
•         google()
•         mavenCentral()
•         gradlePluginPortal()
•     }
• }
• dependencyResolutionManagement {
•     // PREFER_PROJECT allows Capacitor modules to use their own
•     repository definitions
•     repositoriesMode.set(RepositoriesMode.PREFER_PROJECT)
•     repositories {
•         google()
•         mavenCentral()
•         maven { url = uri("https://plugins.gradle.org/m2/") } // ADDED
•         // Ionic libs repository (for Geolocation plugin dependencies)
•         maven { url = uri("https://jitpack.io") } // ADDED
•     }
• }
```

```
• }
```

Add these to the bottom:

```
•  
• // Include Capacitor core module from mcsa-gis-android  
  
• include(":capacitor-android")  
• project(":capacitor-android").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/android/capacitor")  
•  
• // Include Capacitor plugins  
• include(":capacitor-app")  
• project(":capacitor-app").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/app/android")  
•  
• include(":capacitor-file-system")  
• project(":capacitor-file-system").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/file-system/android")  
•  
• include(":capacitor-geolocation")  
• project(":capacitor-geolocation").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/geolocation/android")  
•  
• include(":capacitor-preferences")  
• project(":capacitor-preferences").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/preferences/android")  
•  
• include(":capacitor-share")  
• project(":capacitor-share").projectDir =  
  file("mcsa-gis-android/node_modules/@capacitor/share/android")  
•  
• include(":capacitor-file-picker")
```

```

• project(":capacitor-file-picker").projectDir =
  file("mcsa-gis-android/node_modules/capacitor-file-picker/android")
•
• // Include cordova plugins if any
• include(":capacitor-cordova-android-plugins")
• project(":capacitor-cordova-android-plugins").projectDir =
  file("mcsa-gis-android/android/capacitor-cordova-android-plugins")
•
•

```

Step 2: Update **app/build.gradle.kts**

File Modified:

deal_wf_mcsa-develop/app/build.gradle.kts

Changes Made

1. **Enable MultiDex** (*inside `android { defaultConfig { ... } }`*):

```

defaultConfig {
    // ... existing configs ...

    multiDexEnabled = true
}

```

2. **Disable R8 in debug** (*inside `android { buildTypes { debug { ... } } }`*):

```

buildTypes {
    debug {

```

```
// Disable R8 in debug to prevent plugin loading issues

isMinifyEnabled = false

}

}
```

3. **Add required dependencies** (keep existing ones; add only if missing) (inside *dependencies { ... }*):

```
dependencies {

    // Capacitor core

    implementation(project(":capacitor-android"))

    // Capacitor plugins

    implementation(project(":capacitor-app"))

    implementation(project(":capacitor-file-system"))

    implementation(project(":capacitor-geolocation"))

    implementation(project(":capacitor-preferences"))

    implementation(project(":capacitor-share"))

    implementation(project(":capacitor-file-picker"))

    // Cordova compatibility

    implementation(project(":capacitor-cordova-android-plugins"))

    // Geolocation dependencies
```

```
implementation("io.ionic.libs:iongeolocation-android:1.0.0")

implementation("com.google.android.gms:play-services-location:21.3.0")

implementation("org.jetbrains.kotlinx:kotlinx-coroutines-play-services:1.6.4")


// WebView support

implementation("androidx.webkit:webkit:1.12.1")


// Local tile server

implementation("org.nanohttpd:nanohttpd:2.3.1")


// MultiDex runtime

implementation("androidx.multidex:multidex:2.0.1")

}
```

Reason

Required configuration and dependencies to support **Capacitor core/plugins**, **geolocation services**, **WebView enhancements**, **local tile server functionality**, and **MultiDex** for projects that exceed the Android method limit.

Step 3: Update **AndroidManifest.xml**

File Modified:

deal_wf_mcsa-develop/app/src/main/AndroidManifest.xml

Changes Made

1) Permissions Added

(Add these immediately after the existing `ACCESS_NETWORK_STATE` permission)

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />

<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

2) Permissions Updated

Update `READ_EXTERNAL_STORAGE`

```
<!-- BEFORE -->

<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />


<!-- AFTER -->

<uses-permission

    android:name="android.permission.READ_EXTERNAL_STORAGE"

    android:maxSdkVersion="32" />
```

Update `MANAGE_EXTERNAL_STORAGE`

```
<!-- BEFORE -->

<uses-permission android:name="android.permission.MANAGE_EXTERNAL_STORAGE" />


<!-- AFTER -->

<uses-permission

    android:name="android.permission.MANAGE_EXTERNAL_STORAGE"

    tools:ignore="ScopedStorage" />
```


Update `WRITE_EXTERNAL_STORAGE`

```
<!-- BEFORE -->

<uses-permission

    android:name="android.permission.WRITE_EXTERNAL_STORAGE"

    android:maxSdkVersion="28" />

<!-- AFTER -->

<uses-permission

    android:name="android.permission.WRITE_EXTERNAL_STORAGE"

    android:maxSdkVersion="29" />
```

Already present (no changes required):

- `INTERNET`
- `ACCESS_NETWORK_STATE`
- `READ_MEDIA_IMAGES`, `READ_MEDIA_VIDEO`, `READ_MEDIA_AUDIO`

3) Application Attributes Added

(Add the following attributes to the existing `<application>` tag — place them before `tools:targetApi="31"`)

```
android:requestLegacyExternalStorage="true"

android:usesCleartextTraffic="true"
```

4) Add FileProvider for Capacitor

(Add this provider **after the existing FileProvider** and **before the closing `</application>` tag**)

```
<provider
    android:name="androidx.core.content.FileProvider"
    android:authorities="${applicationId}.fileprovider"
    android:exported="false"
    android:grantUriPermissions="true">

    <meta-data
        android:name="android.support.FILE_PROVIDER_PATHS"
        android:resource="@xml/file_paths" />
</provider>
```

Note: The app already contains a FileProvider using `${applicationId}.provider` for existing functionality. This is an **additional** FileProvider required specifically for Capacitor.

4) Add FileProvider for Capacitor

```
<activity
    android:name=".presentation.ui.activity.HomeActivity"
    android:exported="true"
    android:label="@string/app_name"
    android:theme="@style/Theme.DealMCSA"

    // Added
    android:configChanges="orientation|keyboardHidden|keyboard|screenSize|locale|smallestScreenSize|screenLayout|uiMode|navigation"
/>
```

Note: Why this is needed:

When the user switches system light/dark theme (or other listed configuration changes), Android would

normally destroy and recreate that activity. That tears down and rebuilds the WebView / Capacitor stack and can cause a blank (white) GIS screen, failed asset/tile loading, or duplicate initialization of local services. Declaring these configChanges (especially uiMode) lets the activity handle the change without a full restart, so the GIS session stays stable and predictable during theme and similar configuration updates

Step 4: Create **file_paths.xml** and **variables.gradle**

File Created:

deal_wf_mcsa-develop/app/src/main/res/xml/file_paths.xml

File Content:

```
<?xml version="1.0" encoding="utf-8"?>

<paths>

    <external-path name="external" path="." />

    <external-files-path name="external_files" path="." />

    <cache-path name="cache" path="." />

    <files-path name="files" path="." />

</paths>
```

Reason

This file is required by the Capacitor FileProvider added in Step 3 to define the file system locations that can be safely shared with other apps.

File Created:

deal_wf_mcsa-develop/variables.gradle

```
ext {

    // SDK versions - Matched to client app's existing SDK versions
```

```
minSdkVersion = 26 // Client app uses 26 (Capacitor minimum is 24, so this
is fine)

compileSdkVersion = 35 // Matched to client app's compileSdk

targetSdkVersion = 35 // Matched to client app's targetSdk


// AndroidX versions - these MUST be set here for Capacitor modules

androidxActivityVersion = '1.9.2'

androidxAppCompatVersion = '1.7.0'

androidxCoordinatorLayoutVersion = '1.2.0'

androidxCoreVersion = '1.15.0'

androidxFragmentVersion = '1.8.4'

androidxWebkitVersion = '1.12.1'


// Test versions

junitVersion = '4.13.2'

androidxJUnitVersion = '1.2.1'

androidxEspressoCoreVersion = '3.6.1'


// Cordova/Capacitor

cordovaAndroidVersion = '10.1.1'

kotlin_version = '2.0.21'


// Google Play Services (required by Geolocation plugin)

playServicesLocationVersion = '21.3.0'
```

```
}

// Make these available to all subprojects

subprojects {

    afterEvaluate { project ->

        if (project.hasProperty('android')) {

            project.android {

                compileSdk rootProject.ext.compileSdkVersion

                defaultConfig {

                    minSdk rootProject.ext.minSdkVersion

                    targetSdk rootProject.ext.targetSdkVersion

                }

            }

        }

    }

}
```

Step 5: Copy Custom Plugins from mcsa-gis-android

- 1). MemberActionPlugin.kt
- 2). NativeUploaderPlugin.kt
- 3). OfflineTileServerPlugin.kt
- 4). UdpPlugin.kt
- 5). ZipFolderPlugin.kt
- 6). SessionInfoPlugin.kt

From File location:

`mcsa-gis-android/kt-msca-plugins/`

To file location:

`deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/plugins/`

Step 6: Copy GisCapacitorFragment.kt

From File location:

`mcsa-gis-android/mcsa-fragment-files/`

To file location:

`deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/GisCapacitorFragment.kt`

Step 7: Copy **GisScreen.kt** (Compose Screen)

From File location:

`mcsa-gis-android/mcsa-screen-files/`

To file location:

`deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/presentation/ui/screens/GisScreen.kt`

Step 8: Add **GisScreen** to Navigation

File to Modify:

deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/presentation/ui/screens/HomeScreen.kt

Code to add :

```
AppDestination.Gis -> GisScreen()
```

Step 9: Copy Web Assets (if not ran yarn sync:mcsa-gis-android-sync)

Assets to Copy

- **Web build output**
Copy: mcsa-gis-android/dist/*
To: deal_wf_mcsa-develop/app/src/main/assets/public/
- **Capacitor plugins metadata**
Copy: mcsa-gis-android/android/app/src/main/assets/capacitor.plugins.json
To: deal_wf_mcsa-develop/app/src/main/assets/
- **Capacitor runtime config**
Copy: mcsa-gis-android/android/app/src/main/assets/capacitor.config.json
To: deal_wf_mcsa-develop/app/src/main/assets/

Important

1. These assets must be copied **after building mcsa-gis-android** (yarn build).
2. If mcsa-gis-android is rebuilt, **copy the assets again** to ensure the latest web bundle and Capacitor metadata are included in the app.

Step 10: Additional Required File Changes

File 1: Update BaseActivity.kt

File Modified:

deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/presentation/ui/activity/BaseActivity.kt

Change Required: Update the class declaration.

```
// BEFORE

open class BaseActivity : ComponentActivity()

// AFTER

open class BaseActivity : AppCompatActivity()
```

Reason: Capacitor's Bridge requires AppCompatActivity (which extends FragmentActivity), not ComponentActivity.

File 2: Update themes.xml

File Modified:

```
deal_wf_mcsa-develop/app/src/main/res/values/themes.xml
```

Change Required: Update the theme parent to an AppCompatActivity theme.

```
<style name="Theme.DealMCSA" parent="Theme.AppCompat.Light.NoActionBar" />
```


Reason: Since `BaseActivity` is updated to `AppCompatActivity`, the app theme must inherit from an **AppCompatActivity** parent. This avoids theme/runtime issues that occur when using a **Material** parent theme with `AppCompatActivity`.

File 3: Update `HomeActivity.kt`

File Modified:

`deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/presentation/ui/activity/HomeActivity.kt`

Changes Made (Code)

1) Add these imports (along with existing imports)

```
import android.content.Intent // ADDED (for storage permission)

import android.net.Uri // ADDED (for storage permission)

import android.os.Build // ADDED (for storage permission)

import android.os.Environment // ADDED (for storage permission)

import android.provider.Settings // ADDED (for storage permission)

import androidx.appcompat.app.AlertDialog // ADDED (for storage permission dialog)
```

2) Update `onCreate()` to call storage permission check

```
override fun onCreate(savedInstanceState: Bundle?) {

    super.onCreate(savedInstanceState)

    // ... existing code ...

    checkPermissions()

    checkStoragePermission() // ADDED (call storage permission check)

    // ... rest of existing code ...

}
```

3) Update `checkPermissions()` to include location permissions

```
private fun checkPermissions() {  
  
    val permissions = mutableListOf<String>()  
  
    // Camera, Audio, Bluetooth permissions (existing - unchanged)  
  
    if (ActivityCompat.checkSelfPermission(this, Manifest.permission.CAMERA)  
        != android.content.pm.PackageManager.PERMISSION_GRANTED) {  
  
        permissions.add(Manifest.permission.CAMERA)  
  
    }  
  
    if (ActivityCompat.checkSelfPermission(this, Manifest.permission.RECORD_AUDIO)  
        != android.content.pm.PackageManager.PERMISSION_GRANTED) {  
  
        permissions.add(Manifest.permission.RECORD_AUDIO)  
  
    }  
  
    if (ActivityCompat.checkSelfPermission(this,  
Manifest.permission.BLUETOOTH_CONNECT)  
        != android.content.pm.PackageManager.PERMISSION_GRANTED) {  
  
        permissions.add(Manifest.permission.BLUETOOTH_CONNECT)  
  
    }  
  
    // ADDED: Location permissions (required for Geolocation plugin)  
  
    if (ActivityCompat.checkSelfPermission(this,  
Manifest.permission.ACCESS_FINE_LOCATION)  
        != android.content.pm.PackageManager.PERMISSION_GRANTED) {
```

```

        permissions.add(Manifest.permission.ACCESS_FINE_LOCATION)
    }

    if (ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_COARSE_LOCATION)
        != android.content.pm.PackageManager.PERMISSION_GRANTED) {

        permissions.add(Manifest.permission.ACCESS_COARSE_LOCATION)
    }

    if (permissions.isNotEmpty()) {

        ActivityCompat.requestPermissions(this, permissions.toArray(), 0)
    }
}

```

4) Add storage permission methods (for offline tile server)

```

private fun checkStoragePermission() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.R) {

        if (!Environment.isExternalStorageManager()) {

            showStoragePermissionDialog()

        } else {

            Log.d("HomeActivity", "Storage permission already granted")

        }
    }
}

```

```

    }
}

private fun showStoragePermissionDialog() {
    AlertDialog.Builder(this)
        .setTitle("Storage Permission Required")
        .setMessage(
            "This app needs access to manage all files for the GIS map tile server.\n\n" +
            "Without this permission, the map cannot:\n" +
            "• Load offline map tiles\n" +
            "• Access tile files\n" +
            "• Display the map correctly\n\n" +
            "Please tap 'Open Settings' and enable 'Allow access to manage all files'."
        )
        .setPositiveButton("Open Settings") { _, _ ->
            requestManageStoragePermission()
        }
        .setNegativeButton("Cancel") { dialog, _ ->
            dialog.dismiss()
            Toast.makeText(this, "Map may not work without storage permission",
                Toast.LENGTH_LONG).show()
        }
}

```

```
.setIcon(android.R.drawable.ic_dialog_alert)

.setCancelable(false)

.show()
}

private fun requestManageStoragePermission() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.R) {

        try {

            val intent = Intent(Settings.ACTION_MANAGE_APP_ALL_FILES_ACCESS_PERMISSION)

            intent.data = Uri.parse("package:$packageName")

            startActivity(intent)

        } catch (e: Exception) {

            try {

                val intent = Intent(Settings.ACTION_MANAGE_ALL_FILES_ACCESS_PERMISSION)

                startActivity(intent)

            } catch (e2: Exception) {

                Log.e("HomeActivity", "Failed to open storage settings: ${e2.message}", e2)

                Toast.makeText(

                    this,

                    "Could not open settings. Please grant storage permission manually.",

                    Toast.LENGTH_LONG
```

```
        ).show()

    }

}

}
```

Reason

Runtime permissions are required for:

- **Capacitor Geolocation plugin** (location permissions)
- **Offline GIS tile server** (Android 11+ “Manage All Files” flow to access tile files)

File 4: Update **MCSADealApp.kt**

File Modified:

`deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/MCSADealApp.kt`

1) Add the following imports *(top of file with existing imports)*

```
import androidx.multidex.MultiDex // ADDED

import androidx.multidex.MultiDexApplication // ADDED (if you change the
base class)

import android.app.Application // ADDED (if you keep
Application as base class)

import android.content.Context // ADDED
```

2) Update the application class to install MultiDex

Preferred (if you can change the base class):

```
class MCSADealApp : MultiDexApplication() {  
  
    // ... existing code ...  
  
    override fun attachBaseContext(base: Context) {  
  
        super.attachBaseContext(base)  
  
        MultiDex.install(this)  
  
    }  
}
```

If you must keep extending `Application`:

```
class MCSADealApp : Application() {  
  
    // ... existing code ...  
  
    override fun attachBaseContext(base: Context) {  
  
        super.attachBaseContext(base)  
  
        MultiDex.install(this)  
  
    }  
}
```

Reason

This is required once `multiDexEnabled = true` is enabled in `app/build.gradle.kts`, to ensure the app can load methods across multiple DEX files at runtime.

File 5: Update **proguard-rules.pro**

File Modified:

deal_wf_mcsa-develop/app/proguard-rules.pro

Change Required

Add the following rules at the **end of the file**:

```
# Keep attributes needed for reflection

-keepattributes *Annotation*

-keepattributes Signature

-keepattributes Exceptions

-keepattributes InnerClasses

-keepattributes EnclosingMethod


# Capacitor Plugin Rules - Keep all Capacitor plugins from being obfuscated

-keep class com.capacitorjs.plugins.** { *; }

-keep class com.getcapacitor.** { *; }

-keepclassmembers class * extends com.getcapacitor.Plugin {

    <init>(...);

    *;

}

# Specifically keep GeolocationPlugin and its dependencies
```



```
-keep class com.capacitorjs.plugins.geolocation.** { *; }

-keep class com.capacitorjs.plugins.geolocation.GeolocationPlugin {

    <init>(...);

    *;

}

# Keep all plugin classes referenced in capacitor.plugins.json

-keep class com.capacitorjs.plugins.app.AppPlugin { *; }

-keep class com.capacitorjs.plugins.filesystem.FileSystemPlugin { *; }

-keep class com.capacitorjs.plugins.geolocation.GeolocationPlugin { *; }

-keep class com.capacitorjs.plugins.preferences.PreferencesPlugin { *; }

-keep class com.capacitorjs.plugins.share.SharePlugin { *; }

-keep class com.epicshaggy.filepicker.FilePicker { *; }

# Keep custom plugins

-keep class org.deal.mcsa.plugins.** { *; }

# Keep PluginHandle and Bridge classes that load plugins

-keep class com.getcapacitor.PluginHandle { *; }

-keep class com.getcapacitor.Bridge { *; }

-keep class com.getcapacitor.Bridge$Builder { *; }

-keep class com.getcapacitor.PluginManager { *; }
```

```
# Keep all plugin constructors (critical for reflection-based instantiation)
```

```
-keepclassmembers class com.capacitorjs.plugins.** {
```

```
    public <init>();
```

```
    public <init>(android.content.Context);
```

```
}
```

```
# Keep all public methods in plugins (needed for JavaScript bridge)
```

```
-keepclassmembers class com.capacitorjs.plugins.** {
```

```
    public *;
```

```
}
```

```
# Keep all classes that implement Plugin interface
```

```
-keep class * implements com.getcapacitor.Plugin {
```

```
    <init>(...);
```

```
    *;
```

```
}
```

```
# Don't obfuscate plugin method names (needed for JavaScript bridge)
```

```
-keepclassmembers class com.capacitorjs.plugins.** {
```

```
    @com.getcapacitor.annotation.PluginMethod *;
```

```
}
```

Reason

Prevents R8/ProGuard from obfuscating Capacitor plugin classes and methods, which can otherwise lead to runtime failures such as **"Unable to load plugin instance"** due to reflection-based plugin loading.

File 6: Fix Screenshot to Capture WebView (**HomeScreen.kt**)

File Modified:

deal_wf_mcsa-develop/app/src/main/java/org/deal/mcsa/presentation/ui/screens/HomeScreen.kt

Problem

The screenshot button captures a blank area where the Capacitor WebView should appear. This occurs because WebViews are hardware-accelerated and are not captured using the standard `view.draw(Canvas)` approach.

Solution

Use Android's `PixelCopy` API to capture the actual rendered window surface (equivalent to a system screenshot).

Changes Made (Code)

1) Add Imports

Add the following imports (as indicated, after line 7 / alongside existing imports):

```
import android.graphics.Canvas

import android.view.PixelCopy          // ADDED
import android.os.Handler              // ADDED
import android.os.Looper              // ADDED
import android.view.Window             // ADDED
```

```
import android.util.Log
```

2) Replace **onClick** Handler in **ScreenshotButton** (lines ~424–436)

BEFORE (WebView not captured)

```
onClick = {  
    isVisible = false  
  
    val view = activity.window.decorView.rootView  
  
    val bitmap = createBitmap(activity.window.decorView.width,  
activity.window.decorView.height)  
  
    val canvas = Canvas(bitmap)  
  
    view.draw(canvas) // This does not capture hardware-accelerated WebView  
  
    file.value = saveScreenshotToCache(activity, bitmap)  
  
    isVisible = true  
  
    if (file.value!!.exists()) {  
  
        Log.d("Screenshot", "Saved at: ${file.value!!.absolutePath}")  
  
        showConfirmDialog.value = true  
  
    }  
}
```

AFTER (captures WebView using `PixelCopy`)

```
onClick = {  
    isVisible = false  
  
    val window = activity.window  
    val width = window.decorView.width  
    val height = window.decorView.height  
  
    val bitmap = Bitmap.createBitmap(width, height, Bitmap.Config.ARGB_8888)  
  
    // Use PixelCopy to capture the actual rendered window surface  
    PixelCopy.request(window, bitmap, { result ->  
        if (result == PixelCopy.SUCCESS) {  
            // Successfully captured the window surface (including WebView)  
            file.value = saveScreenshotToCache(activity, bitmap)  
            isVisible = true  
            if (file.value!!.exists()) {
```

```

        Log.d("Screenshot", "Saved at: ${file.value!!.absolutePath}")

        showConfirmDialog.value = true

    }

    } else {

        // Fallback to old method if PixelCopy fails

        Log.e("Screenshot", "PixelCopy failed: $result")

        val view = activity.window.decorView.rootView

        val fallbackBitmap = createBitmap(width, height)

        val canvas = Canvas(fallbackBitmap)

        view.draw(canvas)

        file.value = saveScreenshotToCache(activity, fallbackBitmap)

        isVisible = true

        if (file.value!!.exists()) {

            Log.d("Screenshot", "Saved at: ${file.value!!.absolutePath}")

            showConfirmDialog.value = true

        }

    }

}, Handler(Looper.getMainLooper()))

}

```

Summary of Changes

- **Imports added:** `PixelCopy`, `Handler`, `Looper`, `Window`
- **Screenshot capture updated:** replaced `view.draw(canvas)` with `PixelCopy.request()` to capture the rendered surface
- **Fallback included:** uses the previous capture method if `PixelCopy` fails

Reason

`PixelCopy` captures the actual rendered window surface (similar to a system screenshot), including hardware-accelerated WebView content. This ensures the Capacitor WebView is visible in screenshots.

Note

`PixelCopy` requires Android 7.0+ (API 24+). The fallback ensures compatibility if `PixelCopy` fails on certain devices/configurations.

Final Steps: Build and Test

1. **Sync Gradle**
 - `File` → `Sync Project with Gradle Files`
2. **Build Project**
 - `Build` → `Clean Project`
 - `Build` → `Rebuild Project`
3. **Run App**
 - `Run` → `Run 'app'`

These steps ensure all configuration changes are applied, dependencies are resolved correctly, and the app is built and launched with the updated Capacitor integration.

Important Aspects:

1). New Dev vs Old Dev

a). For the New Dev (dev with no integrated setup):

Follow this document steps completely.

b). For the Old Dev (dev with integrated setup) use the following flow please:

If a new update is received do the step 1 else if taken a pull of integrated setup skip step 1:

Step 1: Replace mcsa-gis-android folder

Delete the old mcsa-gis-android folder

Extract the new mcsa-gis-android.zip to the same location (**deal_wf_mcsa-develop**)

Step 2: Taken a pull integrated branch or any update is in web side only (**Lepton will specify this thing**):

```
cd mcsa-gis-android  
  
yarn install  
  
yarn build  
  
npx cap sync
```

Step 3: Necessary step ** copy assets:

Copy mcsa-gis-android/dist/* → **deal_wf_mcsa-develop**/app/src/main/assets/public/

Copy mcsa-gis-android/android/app/src/main/assets/capacitor.plugins.json →
deal_wf_mcsa-develop/app/src/main/assets/

Copy mcsa-gis-android/android/app/src/main/assets/capacitor.config.json →
deal_wf_mcsa-develop/app/src/main/assets/

No need to copy plugins or fragment files if only web code changed.

***** But if the change is in the native code (LEPTON will specify that which file was changed with path in the mcsa-gis-android) *****

2). If new plugin is added by lepton team:

***** Make sure to do these steps *****

Please first unzip the new mcsa-gis-android folder and please replace it in the **deal_wf_mcsa-develop**

Run these command first :

```
yarn install  
yarn build  
npx cap sync
```

Copy mcsa-gis-android/dist/* → **deal_wf_mcsa-develop**/app/src/main/assets/public/

Copy mcsa-gis-android/android/app/src/main/assets/capacitor.plugins.json → **deal_wf_mcsa-develop**/app/src/main/assets/

Copy mcsa-gis-android/android/app/src/main/assets/capacitor.config.json → **deal_wf_mcsa-develop**/app/src/main/assets/

You will definitely receive a new [GisCapacitorFragement.kt](#) and new Plugin Code as well